

CHAPTER – 8

ARRAY:

- Array is a collection similar data type elements.
- Eg: `int [] arr={1,2,3,4,5};`

KEY FEATURES:

- Array index starts with 0
- Array size is fixed
- Array elements must be same data type

SYNTAX:

`dataType[] arrayName = new dataType[size];`

ADVANTAGES:

- Code Simplicity
- Easy Access
- Iteration
- Memory Efficiency
- Sorting / Searching

TYPES OF ARRAY:

- One – Dimensional Array
- Two – Dimensional Array

ONE – DIMENSIONAL ARRAY:

- Single row of elements.

EXAMPLE:

```
public class OneDimArr {  
    public static void main(String [] args) {  
        int [] arr= {10,20,30,40};  
        for(int i=0; i<arr.length; i++) {  
            System.out.println("Numbers in Array: "+arr[i]);  
        }  
    }  
}
```

TWO – DIMENSIONAL ARRAY:

- An array of arrays – used to represent tables.

EXAMPLE:

```
public class TwoDimArr {  
    public static void main(String [] args) {  
        int [][] arr= {  
            {10,20},  
            {30,40},  
            {50,60}  
        };  
        for (int i=0; i<arr.length; i++) {  
            for (int j=0; j<arr[i].length; j++) {  
                System.out.print(arr[i][j]+" ");  
            }  
        }  
    }  
}
```

```

        }
        System.out.println();
    }
}

```

JAGGED ARRAY:

- Jagged array is a two dimensional array where each row can have a different number of columns.

EXAMPLE:

```

public class JaggedArrayExample {
    public static void main(String[] args) {
        int[][] arr = new int[3][];
        arr[0] = new int[] {10, 20};
        arr[1] = new int[] {30, 40, 50};
        arr[2] = new int[] {60, 70, 80, 90};
        System.out.println("Jagged Array Elements:");
        for (int i = 0; i < arr.length; i++) {
            for (int j = 0; j < arr[i].length; j++) {
                System.out.print(arr[i][j] + " ");
            }
            System.out.println();
        }
    }
}

```

